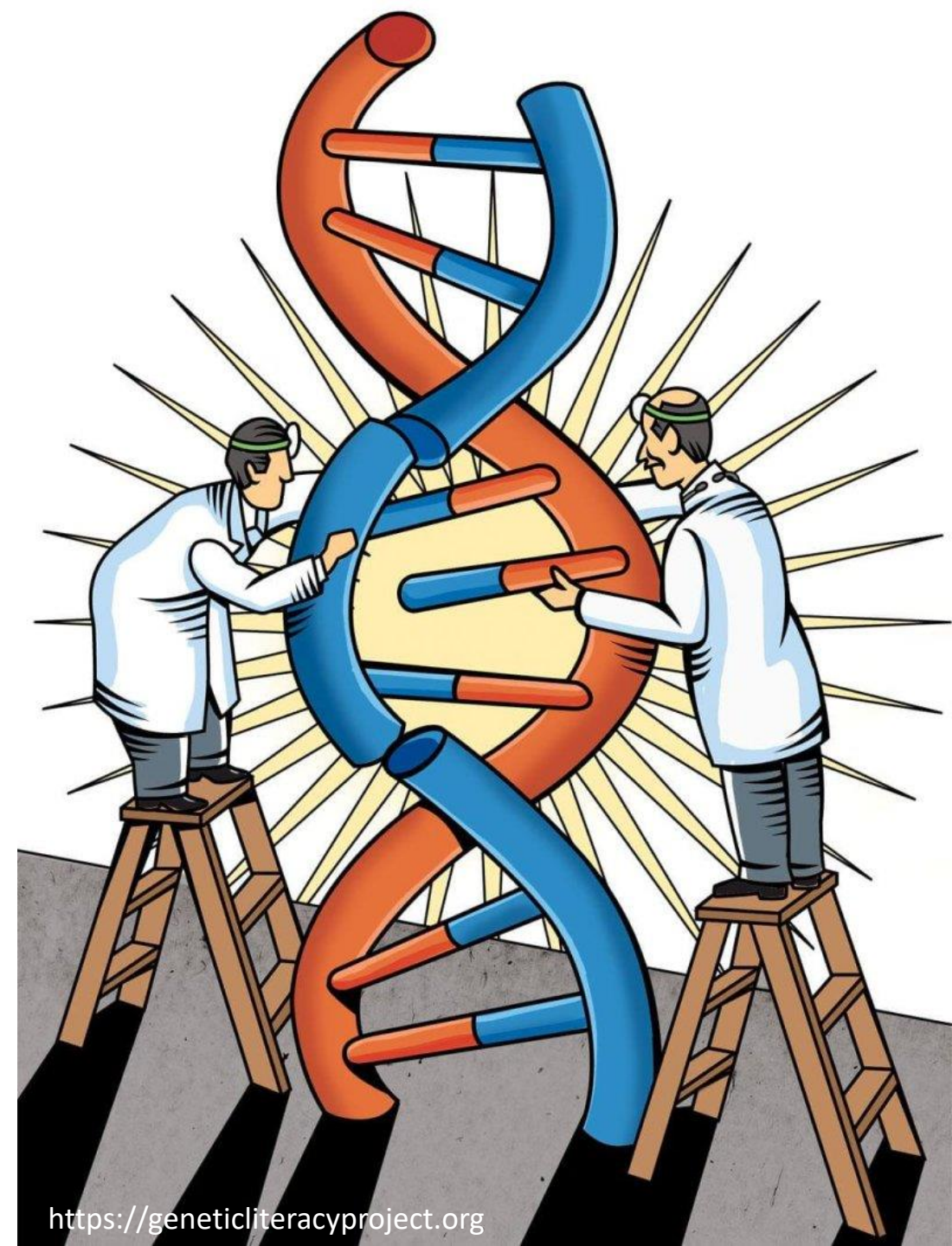


CRISPR in action

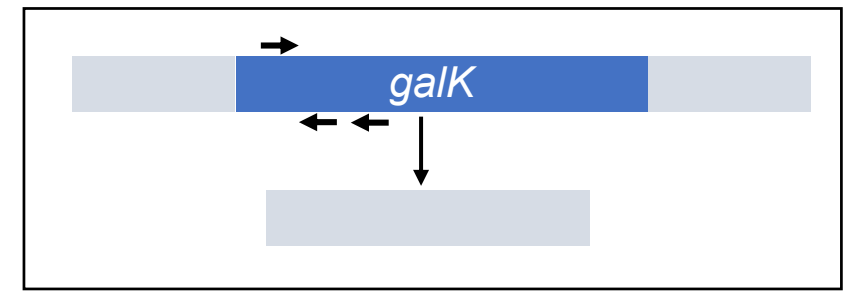
practice and theory of genome engineering

Day 5
August 21, 2025

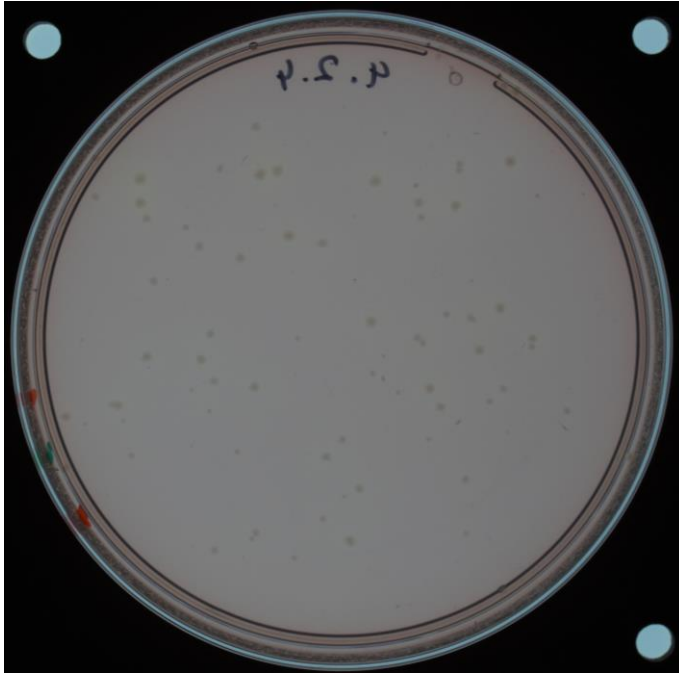
Sibylle Vonesch
Laboratory for Genome Editing and
Systems Genetics



Whole gene deletion (Experiment 1)

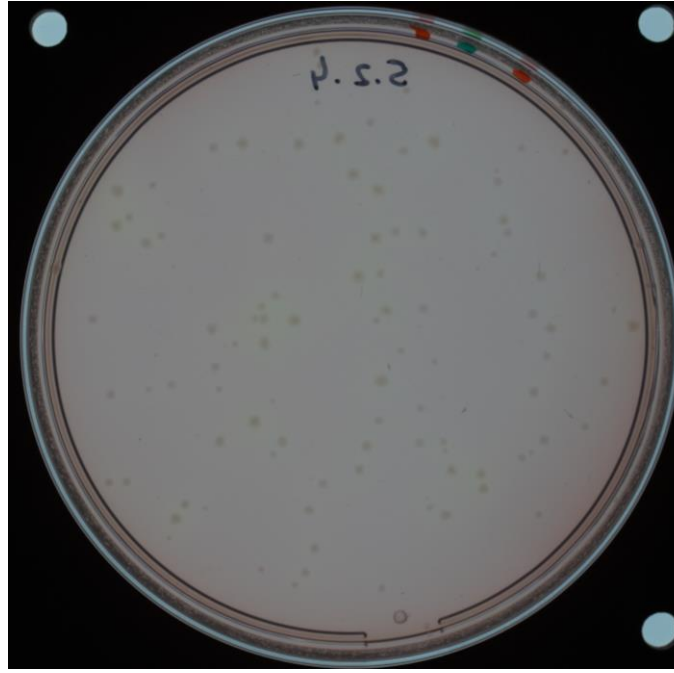


gRNA1



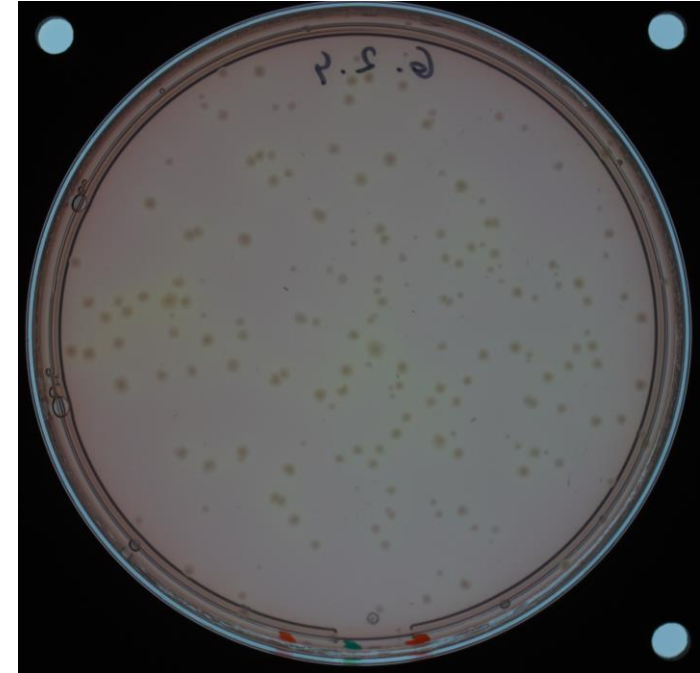
Off-target score 100
50% GC
PAM disruption
75.7 on target score
Self-complementarity 0

gRNA2

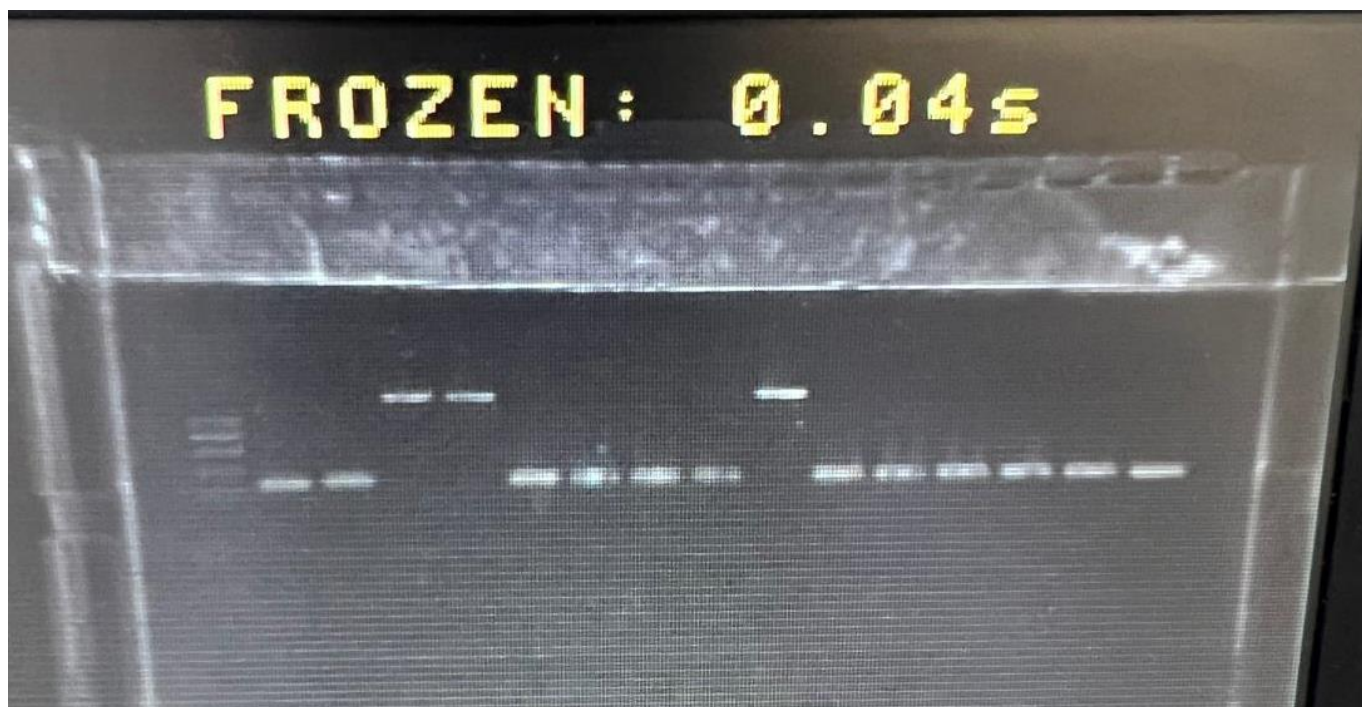


Off-target score 100
60% GC
PAM distance 4
74.5 on target score
Self-complementarity 0

gRNA3



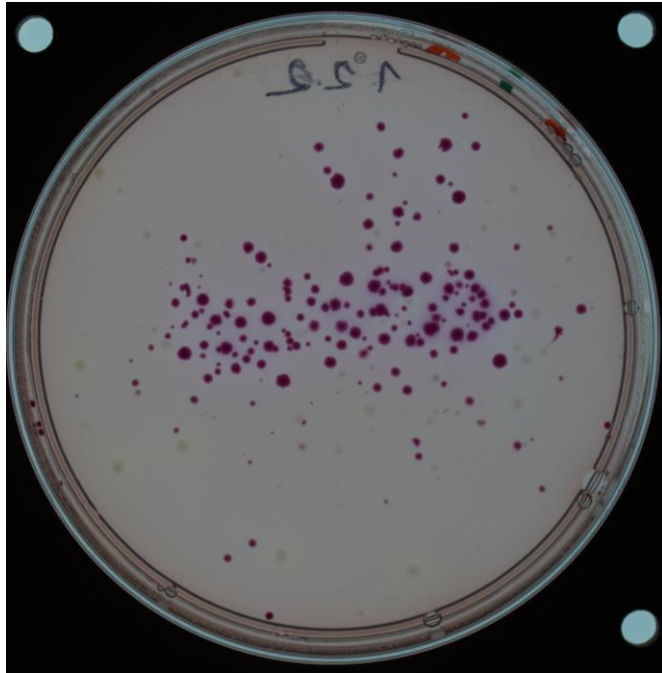
Off-target score 100
55% GC
PAM disruption
71.2 on target score
Self-complementarity 0



WT	1526 bp
galk KO	200 bp

PTC (Experiment 2)

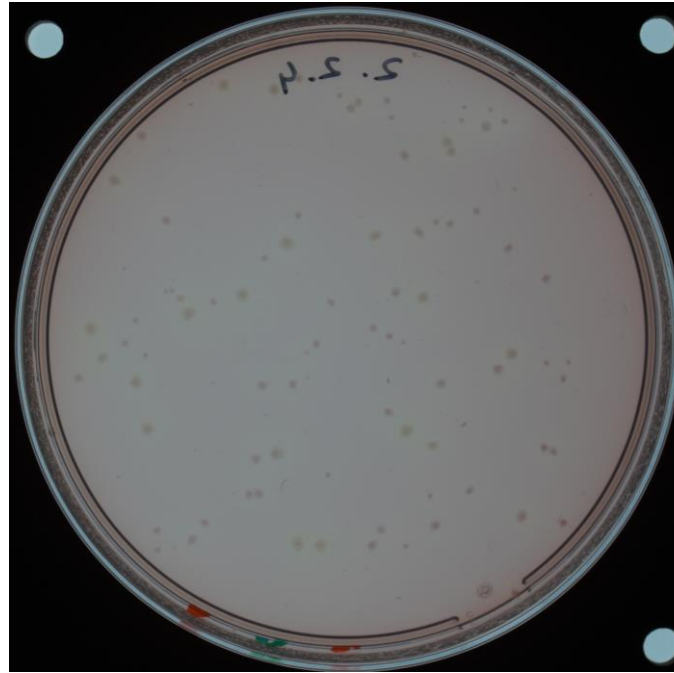
gRNA1



Off-target score 100
50% GC
PAM disruption
75.7 on target score
Self-complementarity 0

1:100 dilution

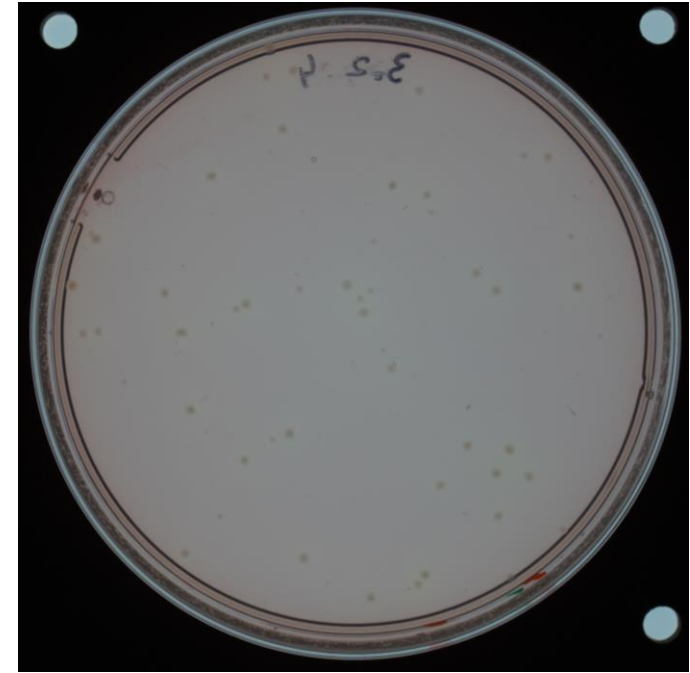
gRNA2



Off-target score 100
60% GC
PAM distance 4
74.5 on target score
Self-complementarity 0

1:10,000 dilution

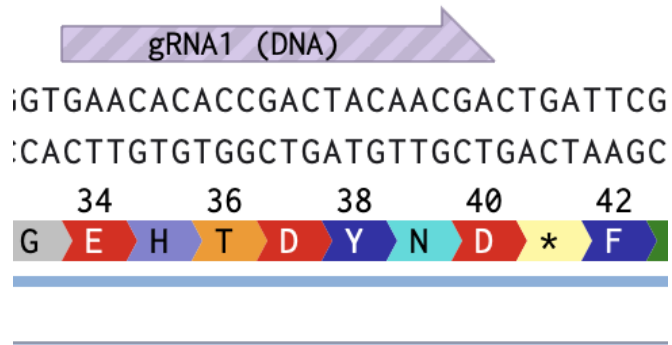
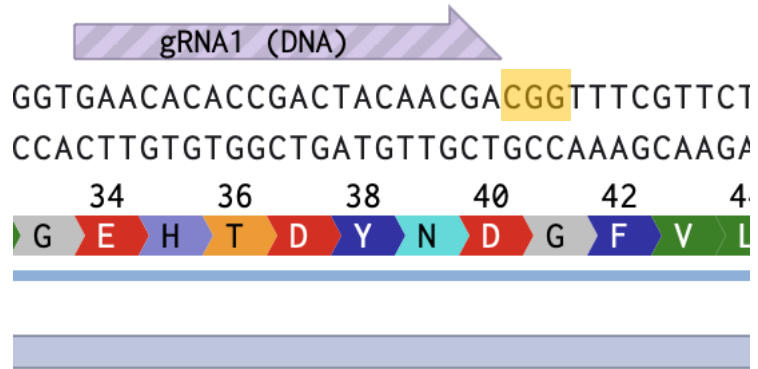
gRNA3



Off-target score 100
55% GC
PAM disruption
71.2 on target score
Self-complementarity 0

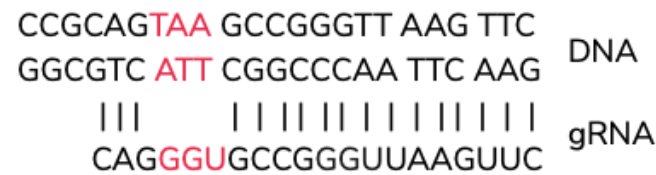
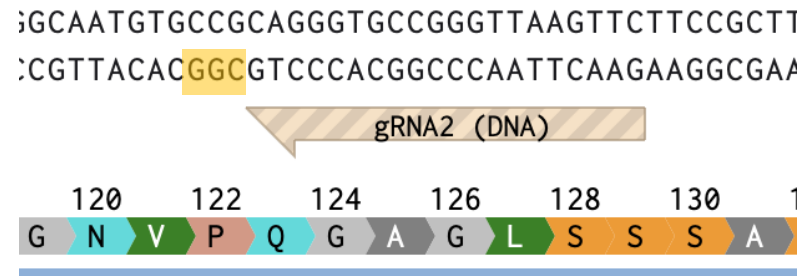
1:10,000 dilution

G41* (GGT to TGA)



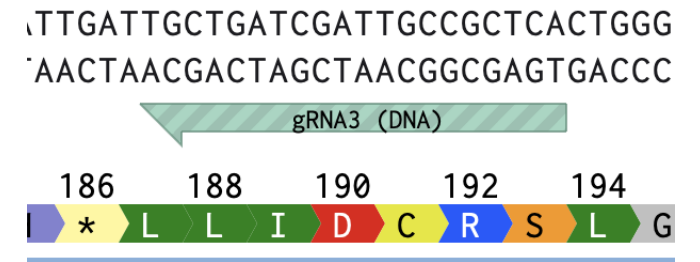
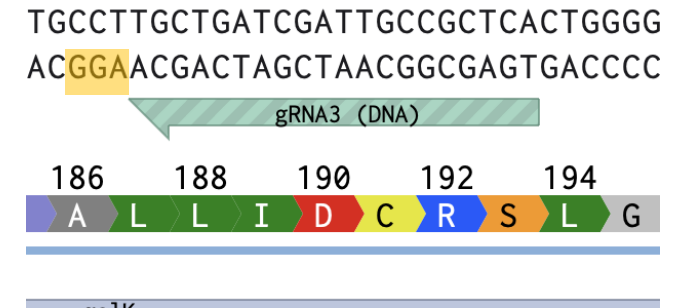
No PAM in edited target
 – no cleavage

G124* (GGT to TAA)



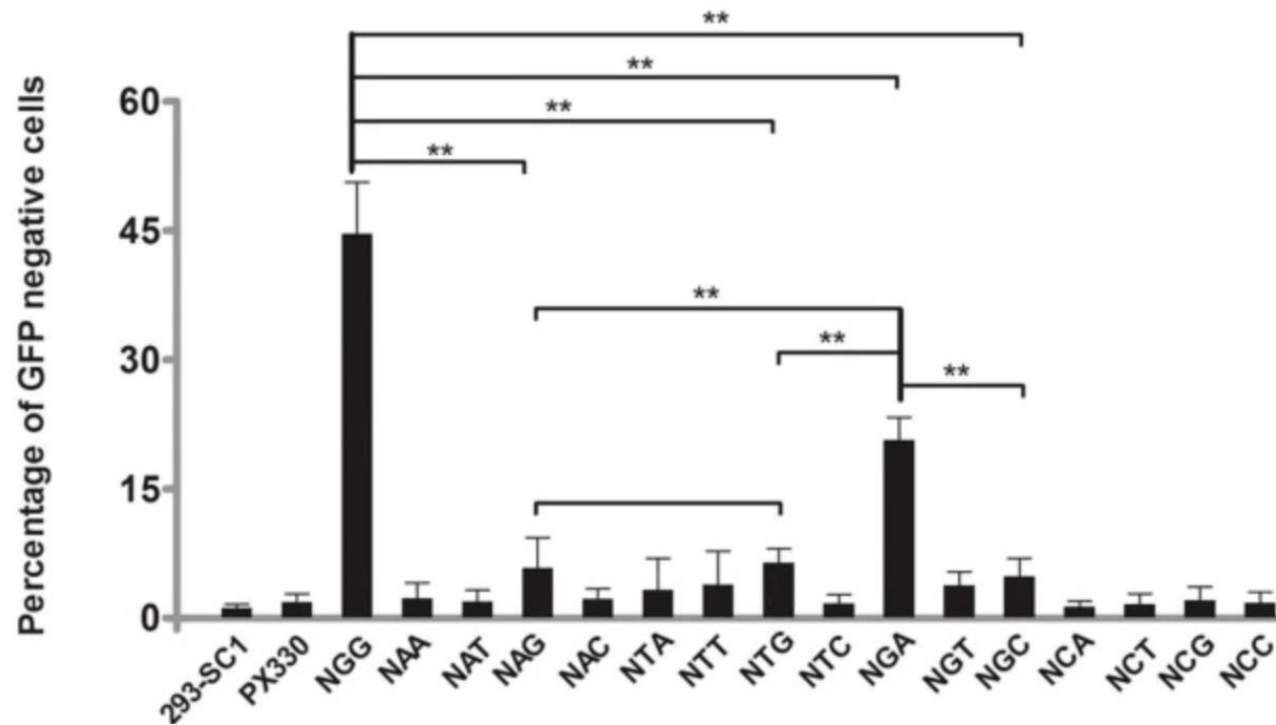
3 bp mismatch to edited target
 – no cleavage

A186* (GCC to TGA)

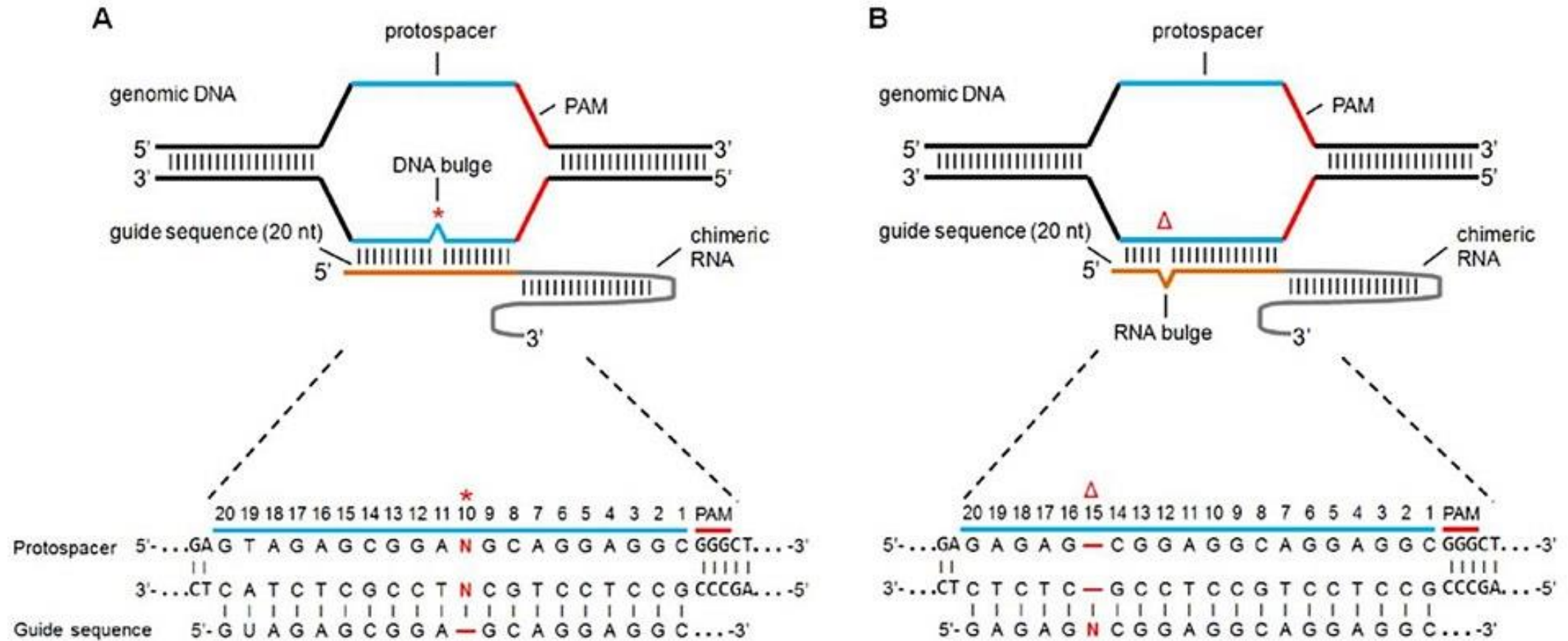


No PAM in edited target
 – no cleavage

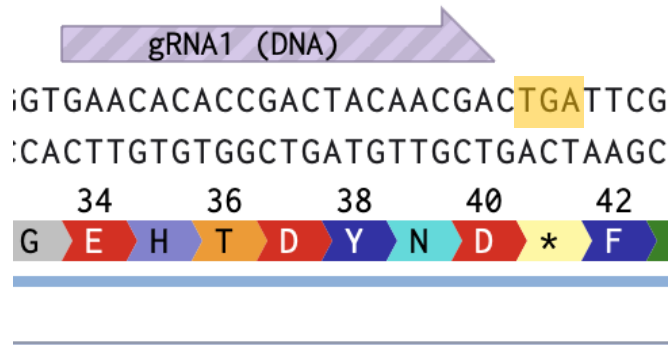
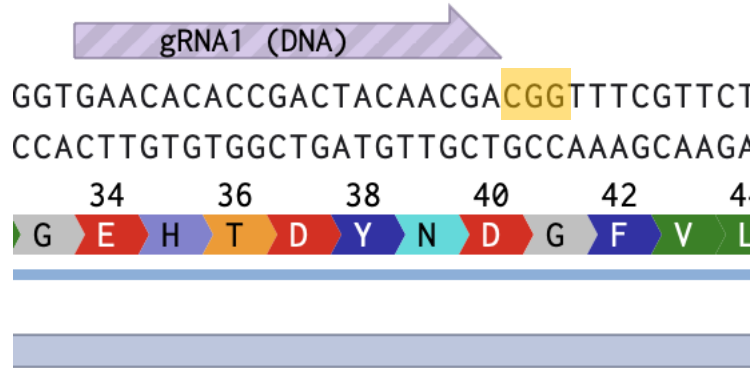
Noncanonical PAMs for SpCas9



SpCas9 sometimes tolerates small DNA and RNA bulges

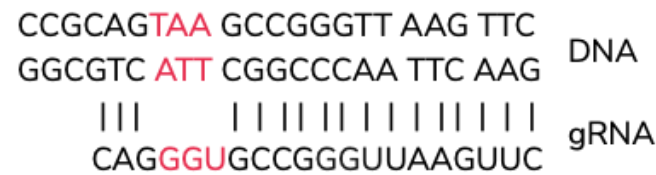
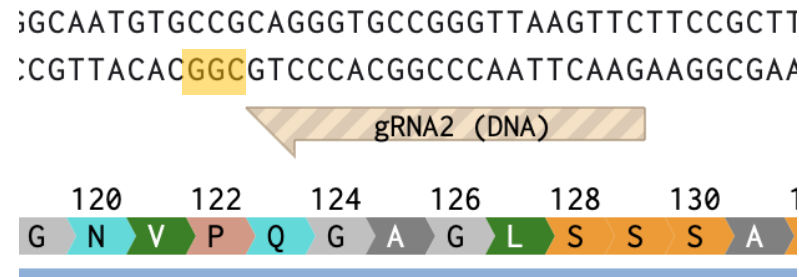


G41* (GGT to TGA)



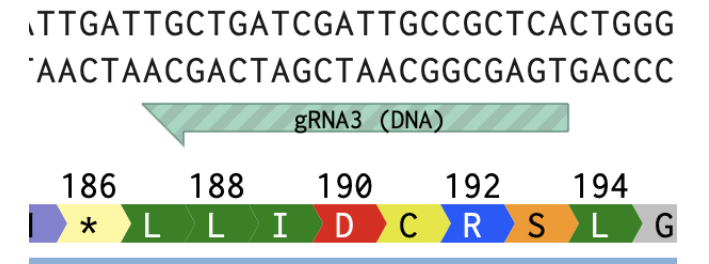
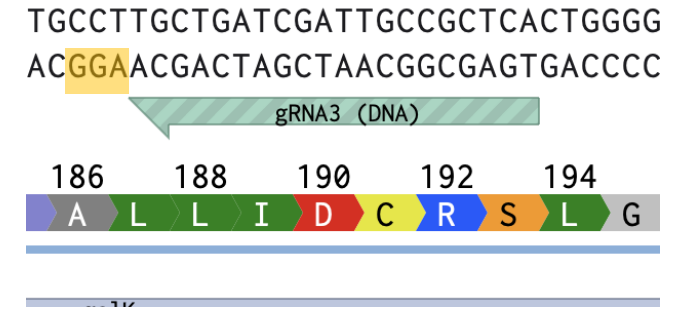
DNA bulge + noncanonical
PAM in edited target
– sometimes cleavage

G124* (GGT to TAA)



3 bp mismatch to edited target
– no cleavage

A186* (GCC to TGA)



No PAM in edited target
– no cleavage

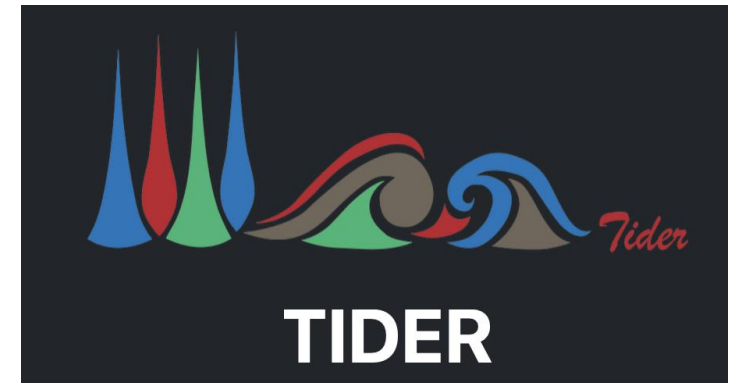
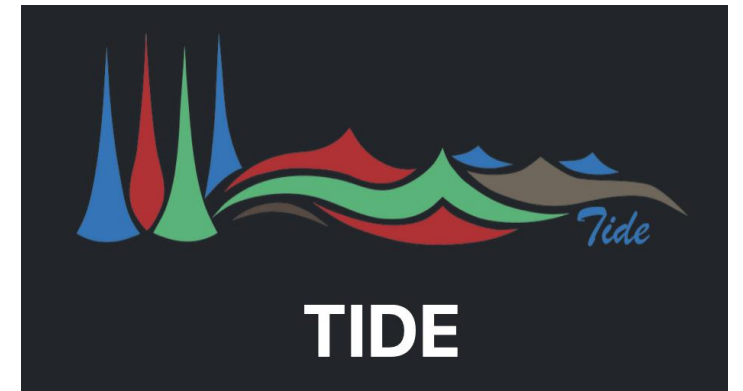
Practical: Tools for analysing editing outcomes

TIDE – indel decomposition from **Sanger** data

TIDER – decomposition of indels and templated (HDR) edits from **Sanger** data

<https://tide.nki.nl>

Use TIDE example data (check box)



CRISPResso2 – super easy **NGS** amplicon sequencing analysis -

<https://crispresso2.pinellolab.org/submission>

Exercises/day5



CRISPResso2

Analysis of genome editing outcomes from deep sequencing data

Improving HDR editing

Experimental design and design of components

Already discussed:

- gRNA design: place the cut site as close as possible to the intended change, ensure the edit disrupts the PAM or gRNA-DNA hybridization region.
- Adjust homology arms to application – longer insertions require longer HAs, some species require longer HAs than others, asymmetry & strand choice can help with ssODNs.
- Phosphorothioate linkages for ssODNs to protect from exonucleases.

Improving HDR editing in *E. coli* (and other bacteria)

λ Red System (Recombineering)

Origin: Derived from bacteriophage λ; enables homologous recombination in *E. coli*.

(also other phage recombinases)

Key proteins:

- **Exo** – 5'→3' exonuclease; processes linear dsDNA to create ssDNA overhangs.
- **Beta** – binds ssDNA; promotes annealing to complementary chromosomal DNA.
- **Gam** – inhibits RecBCD nuclease; protects linear DNA from degradation.

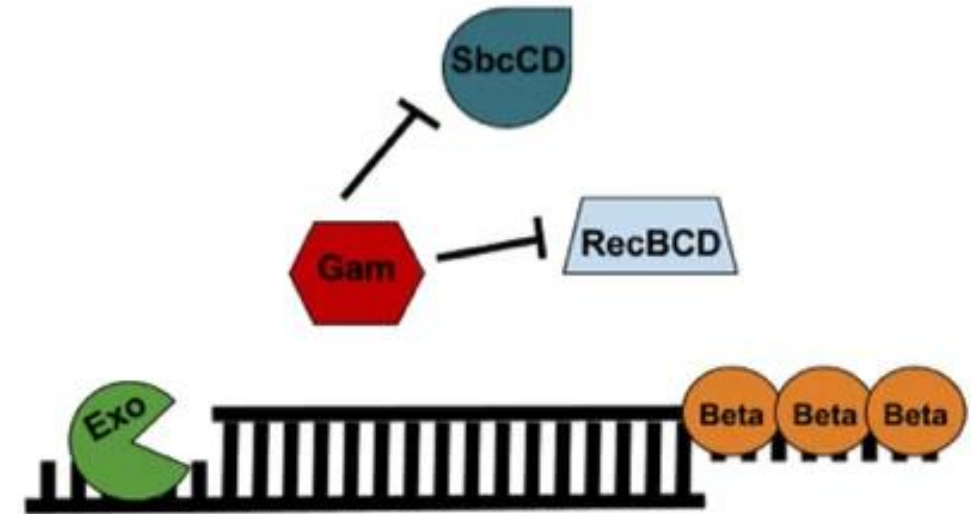
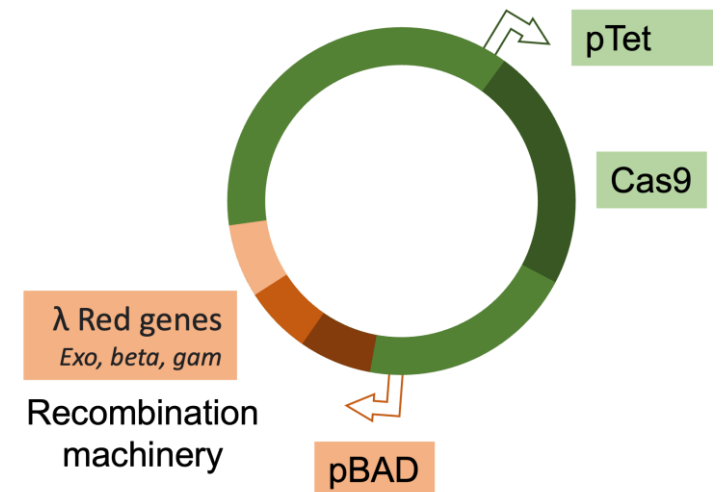


Figure from Addgene blog



Eukaryotes

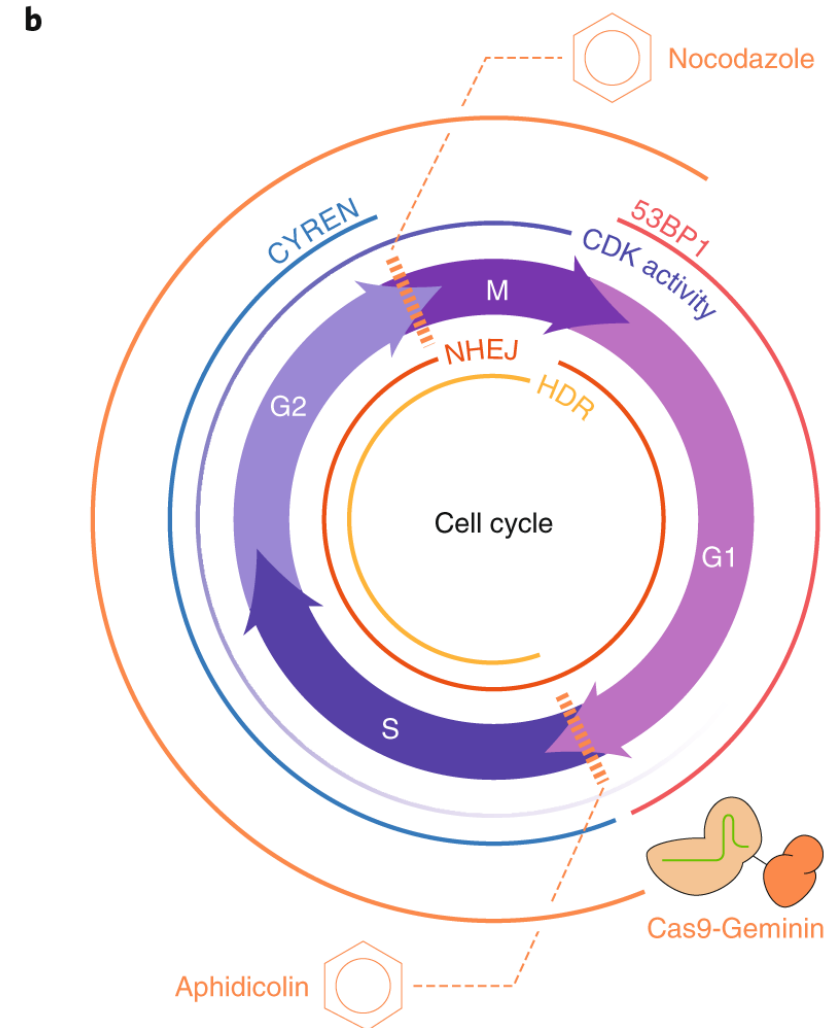
Timing & cell cycle control

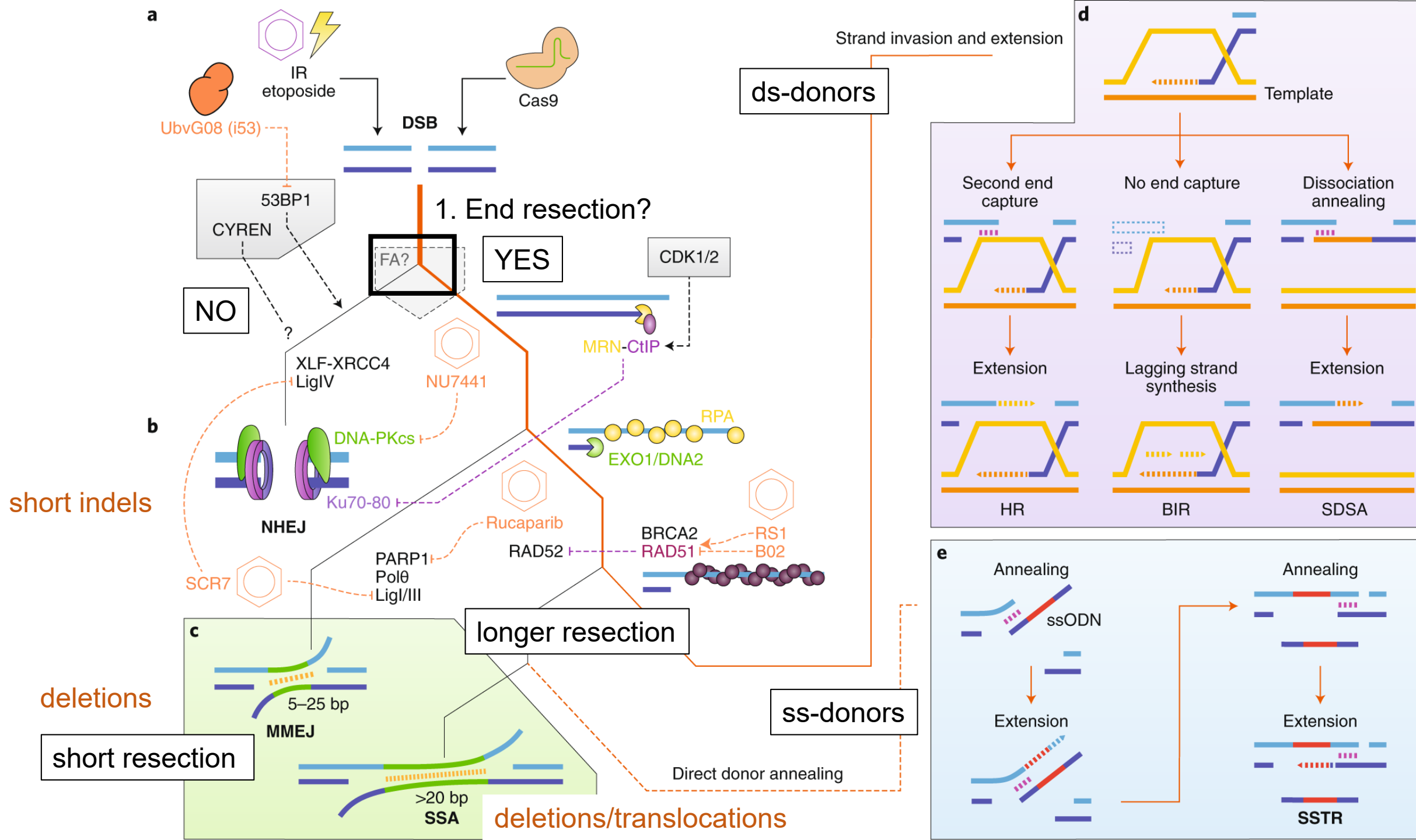
HDR is most active in **S/G2 phase**, so shifting Cas9 activity to those stages can help:

Cell synchronization - thymidine, nocodazole to enrich cells in S/G2 before delivery.

Cell cycle-restricted Cas9 - fusing Cas9 to a degron (e.g. Geminin) so it's degraded in G1 but active in S/G2.

In *S. cerevisiae*, other yeasts: transform plasmid as two linear fragments with homology overlaps





Impairing competing repair pathways

Permanent changes – KO of NHEJ proteins

- Ku70/Ku80
- Lig IV
- DNA PKcs

Impairing competing repair pathways

Permanent changes – KO of NHEJ proteins

- Ku70/Ku80
- Lig IV
- DNA PKcs

Chemical inhibitors:

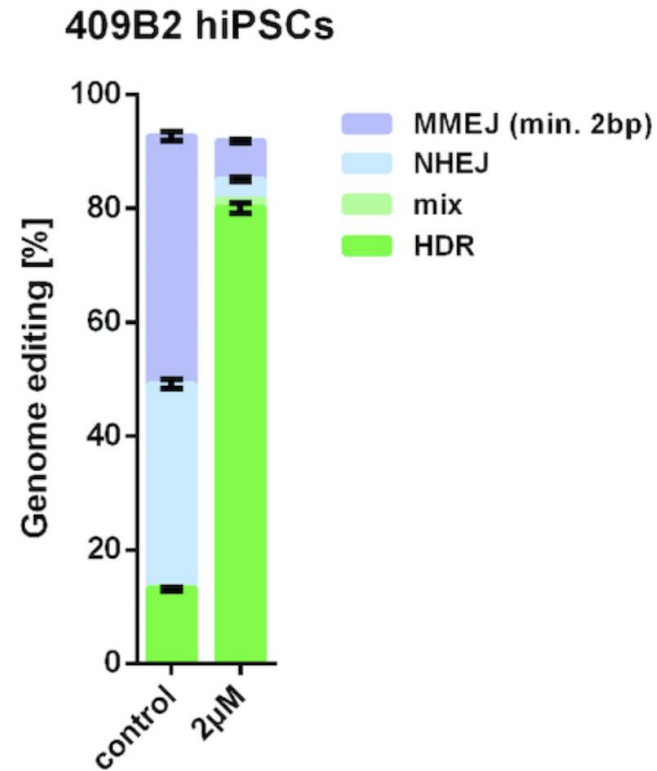
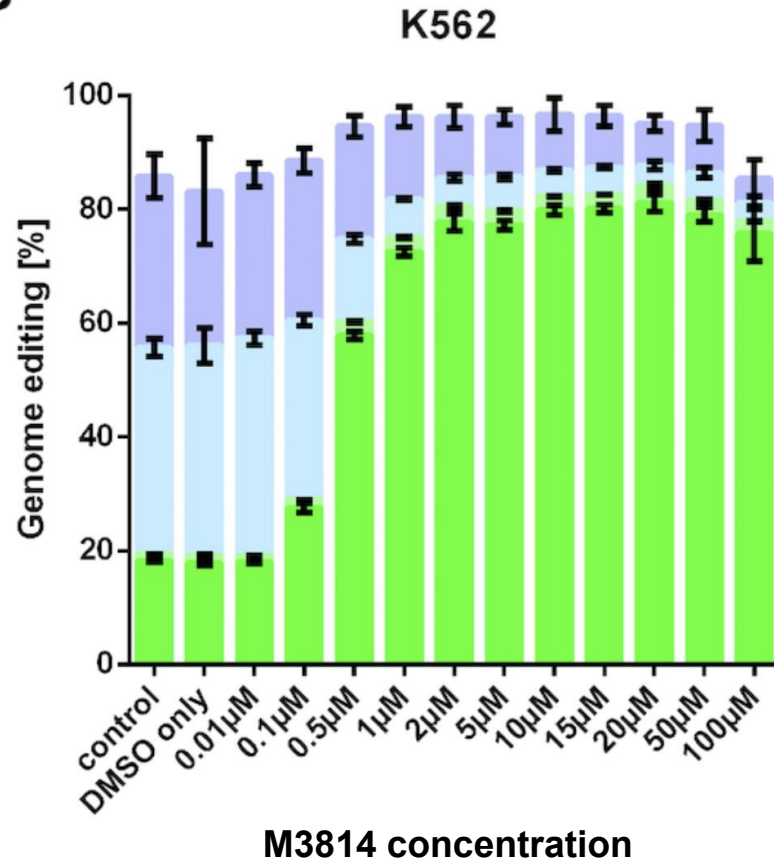
DNA ligase inhibitors (SCR7; DNA ligase IV, Lig I/III) - mixed results in literature but can help in some lines.

DNA-PKcs inhibitors (NU7441, M3814, AZD7648) - block NHEJ initiation, often more consistent than SCR7. Not all eukaryotes have a true ortholog.

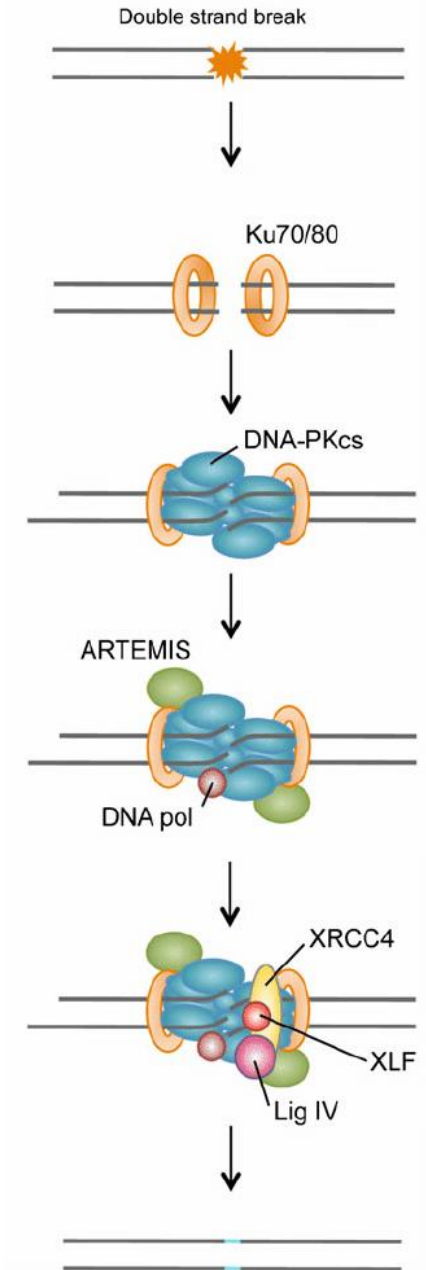
Pol θ inhibitors (novobiocin, ART558) - suppress MMEJ, shifting repair toward HDR.

Inhibiting NHEJ with drugs

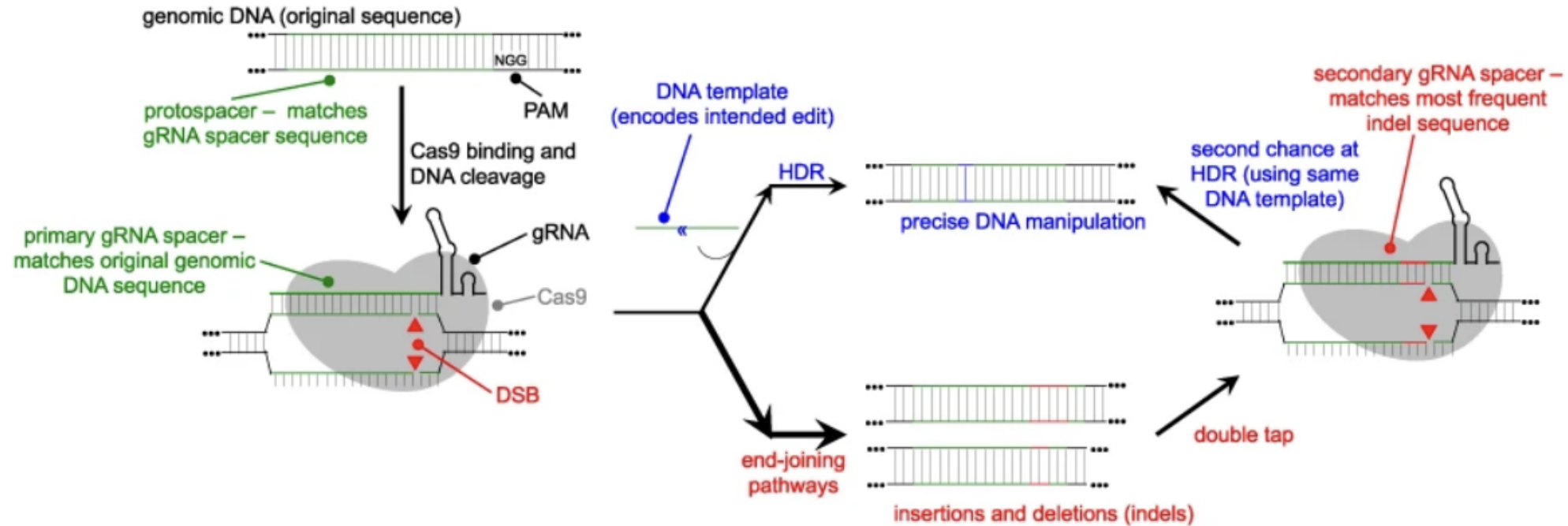
B



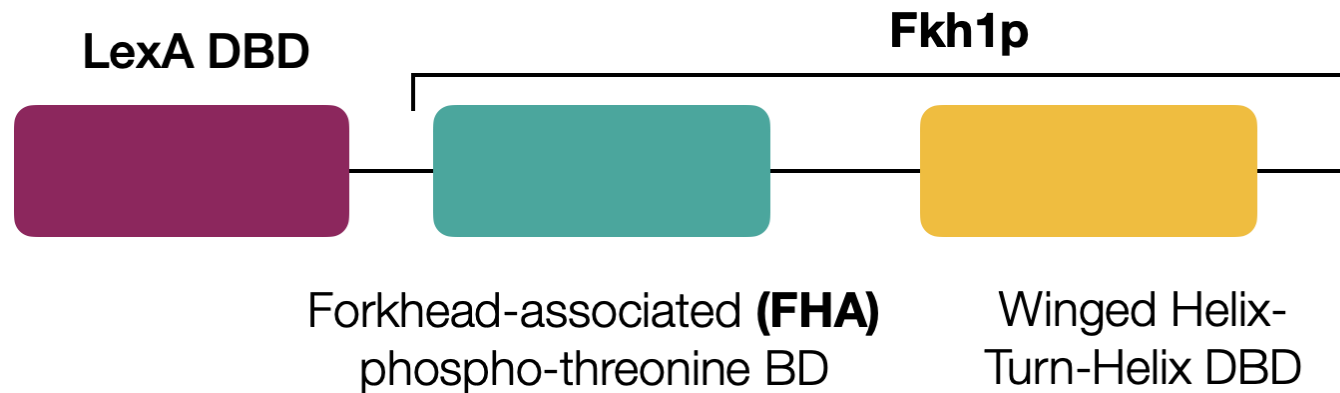
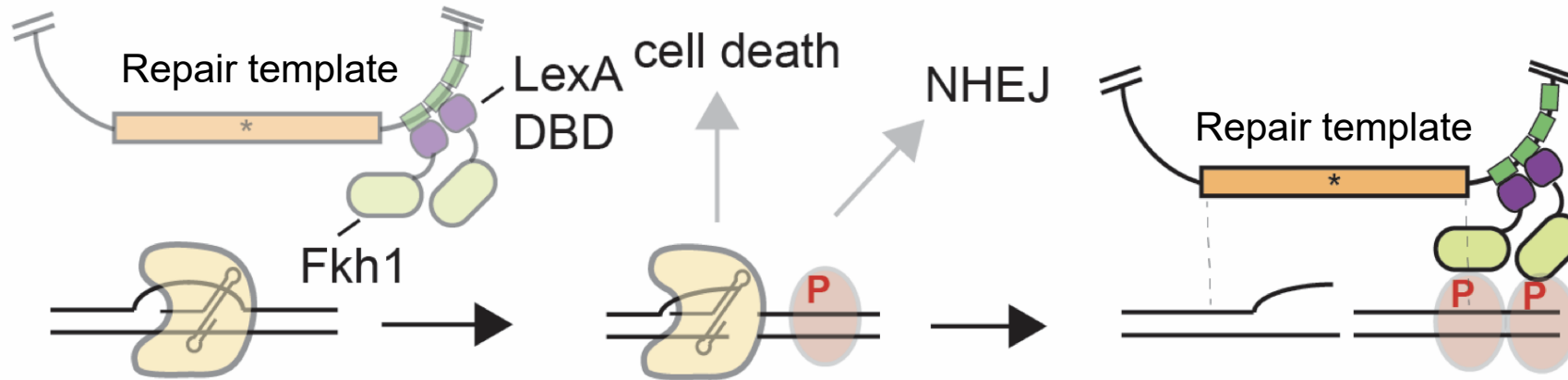
Non-homologous end joining



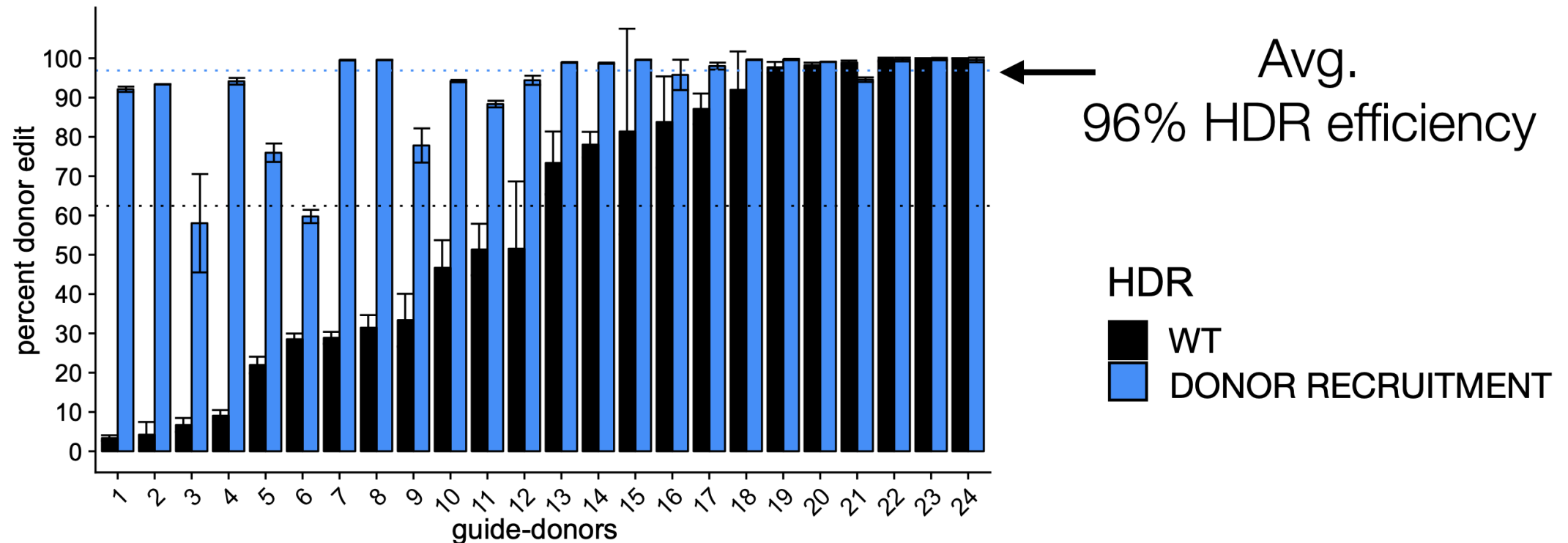
Targeting main indel byproduct



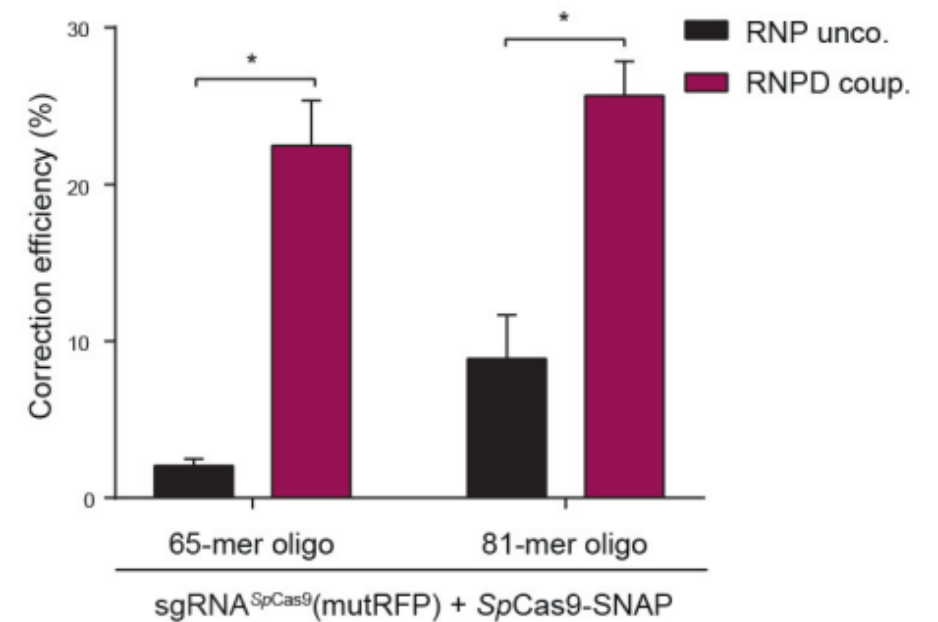
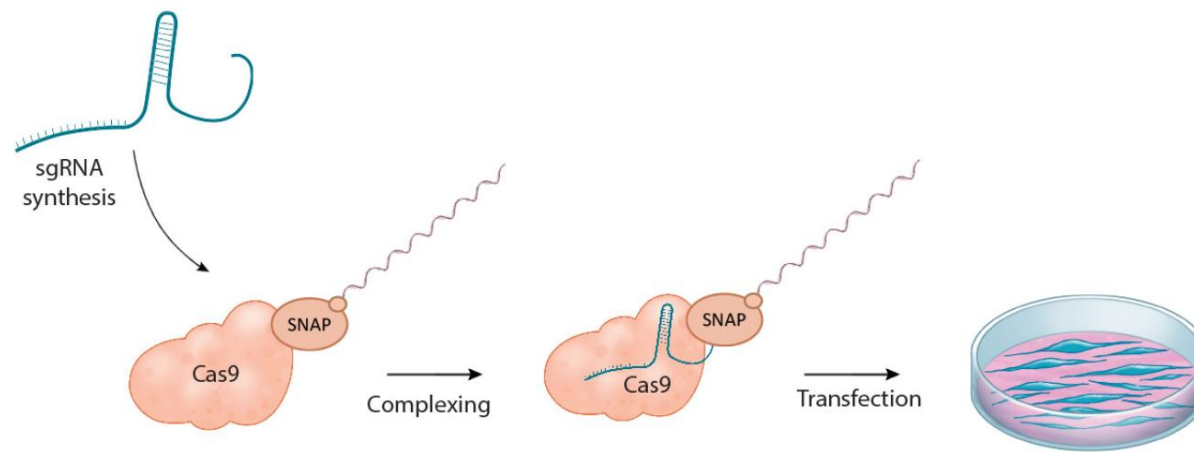
Improving HDR by recruiting donor to sites of DNA breaks



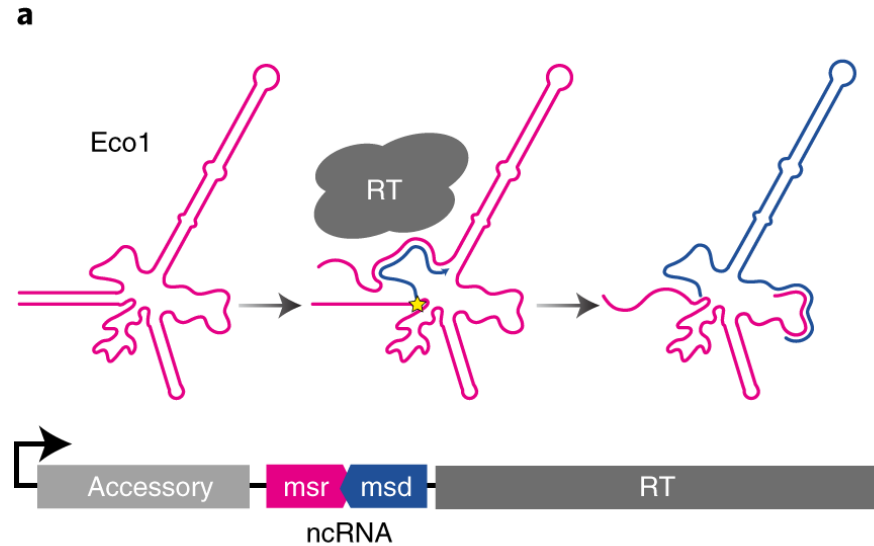
Improving HDR by recruiting donor to sites of DNA breaks



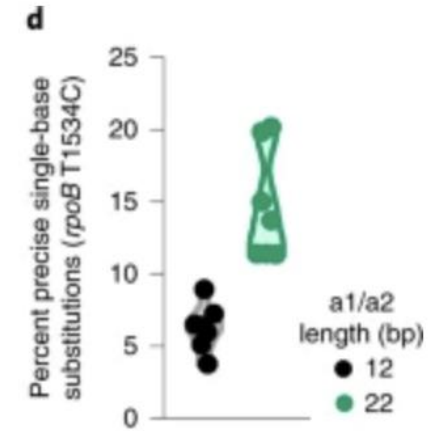
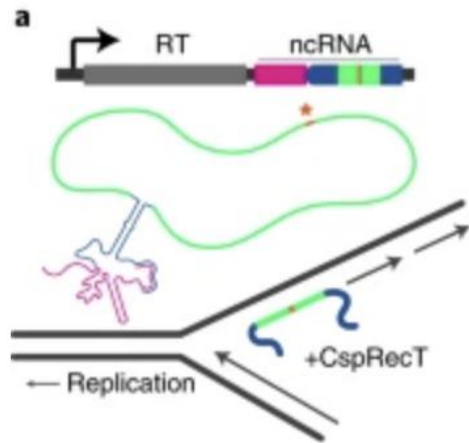
Enrichment of donor at ds-breaks via covalent attachment



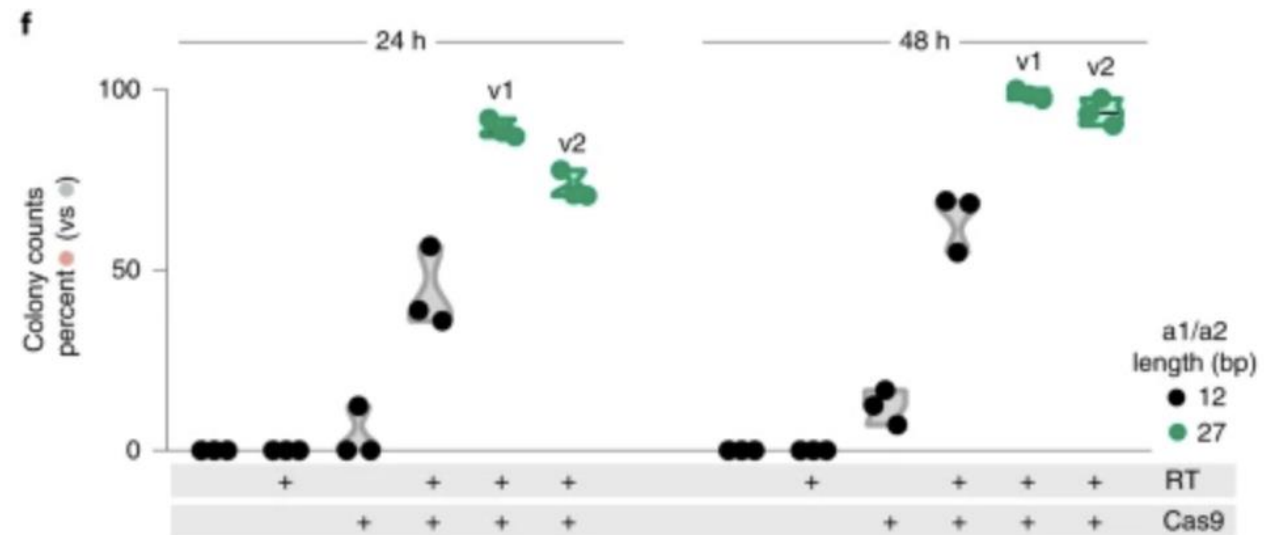
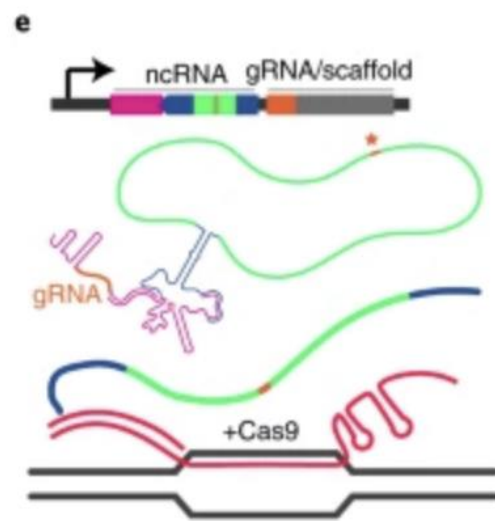
Increasing intracellular donor concentration



- **Retrons:** reverse transcriptase (RT) and ncRNA (msr-msd) that is both the primer and template for RT (plus accessory proteins)
- RT partially reverse transcribes ncRNA → single-stranded DNA with characteristic hairpin structure
- Only msd region of ncRNA is reverse transcribed → donor placement in msd

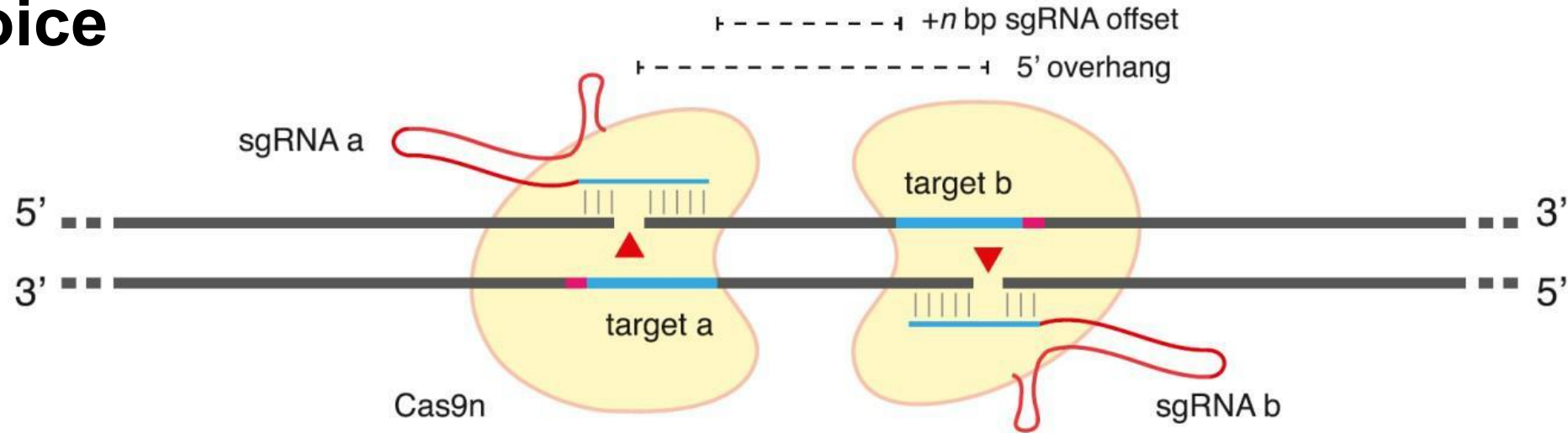


E. coli



S. cerevisiae

Cas variant choice



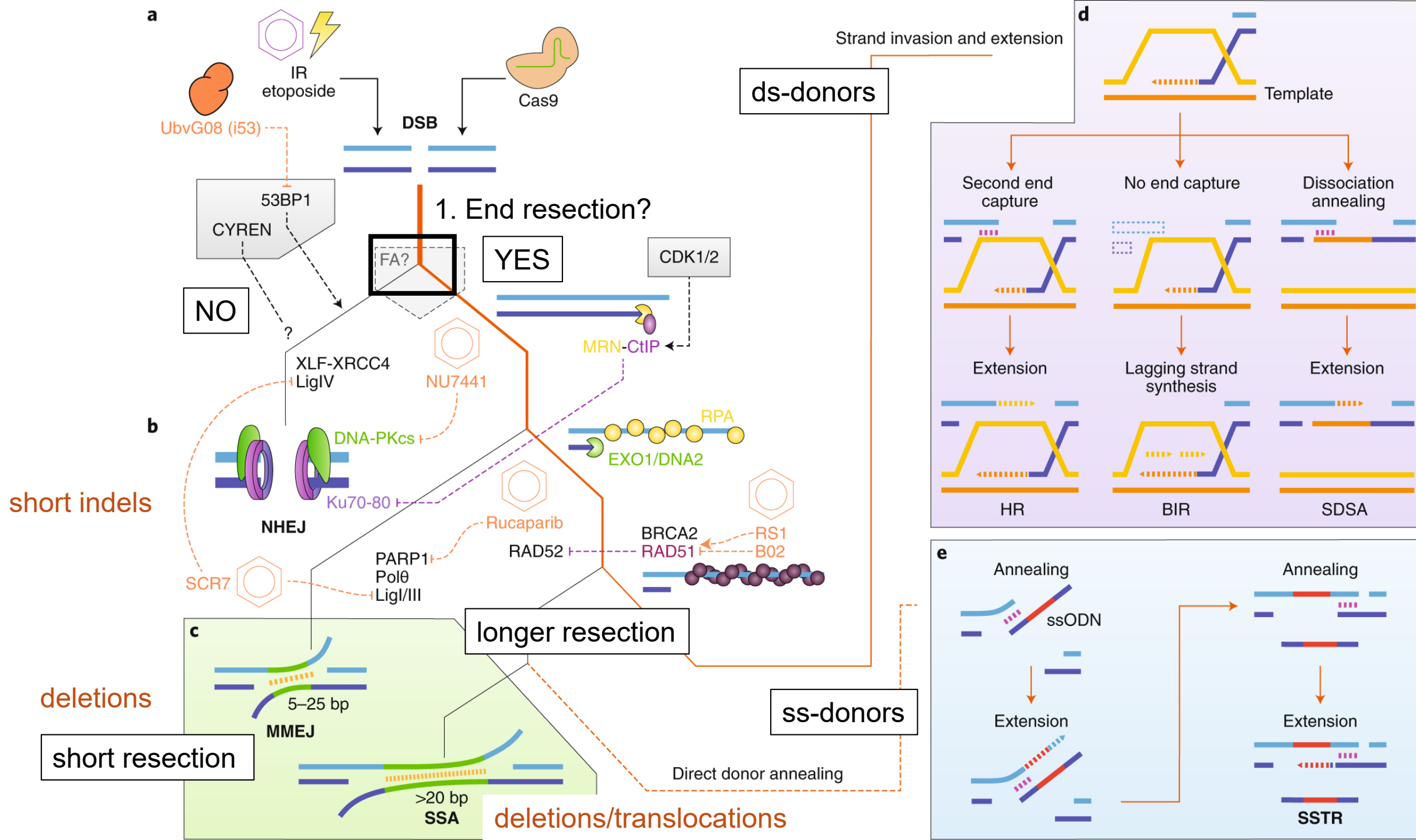
Cas9 nickase paired with two guides (nCas9) - creates staggered nicks, increasing HDR:indel ratio.

- nCas9 D10A cleaves target strand (inactive RuvC domain)
- nCas9 H840A cleaves non-target strand (inactive HNH domain)

Cas12a (Cpf1) - produces staggered overhangs and can sometimes favor HDR.

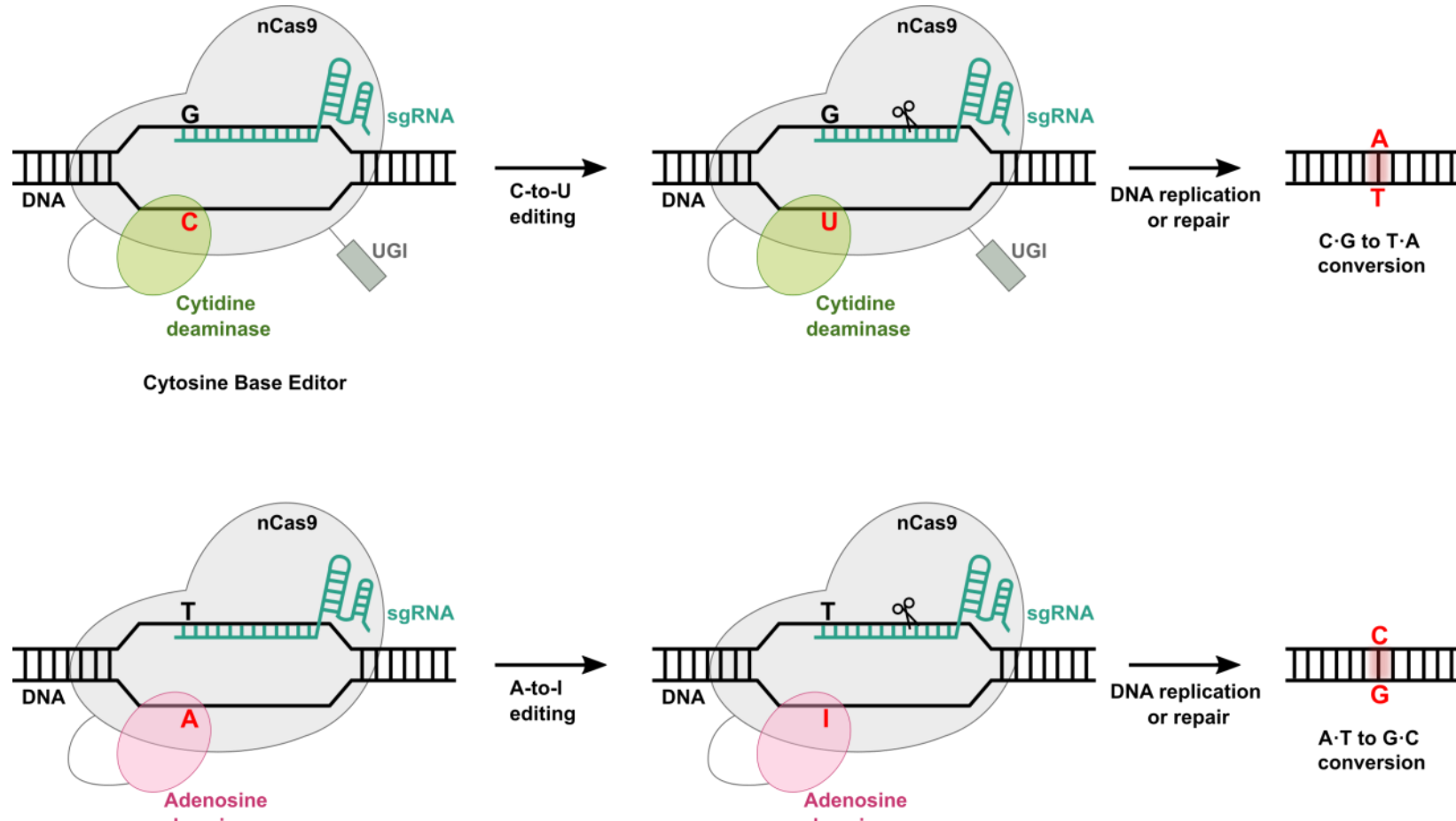
Efficacy? More 'picky' in terms of PAM – NTTTV -> DTTTV for highest efficacy^{1,2}

Cas9 fused to HDR factors Cas9-CtIP, Cas9-RAD52, Cas9-MRN complex components – very modest improvements in general



Alternative editing systems

Base editing

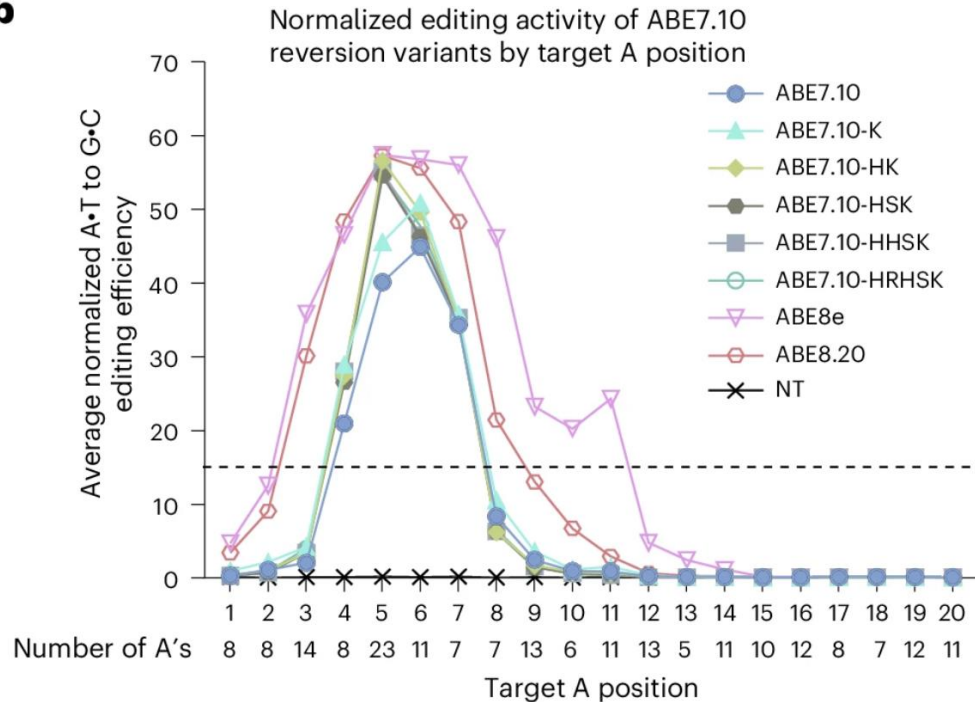


- No dependence on HDR and no DSBs
- Currently your best bet for precise edits in plants

Main issues

- Bystander editing

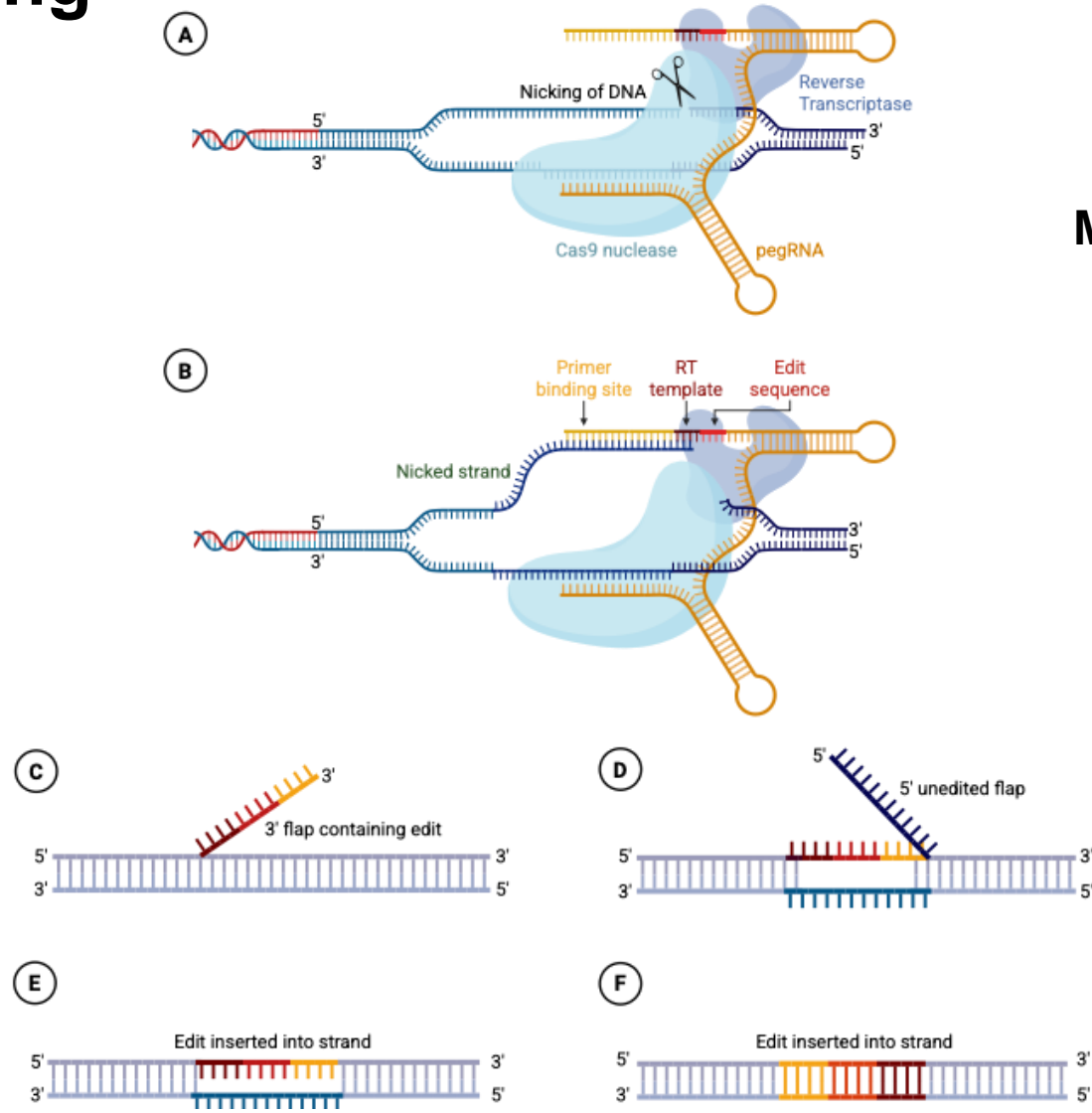
b



<https://doi.org/10.1038/s41587-026-03045-z>

- Cas-independent off targets: deaminase ssDNA
- Edit diversity limited (A to G, C to T)
- PAM dependence is more limiting: A or C needs to be in correct position in editing window

Prime editing



Main issues:

- Often needs substantial optimization in new organism
- Efficacy typically lower & more variable than BE, HDR
- large

No dependence on HDR and (sometimes) no DSBs



Review

From edits to insights: precision microbial engineering for systems biology

Vlad S Băţăgui^{1 2}*, Antoine Delhaye^{1 2}*, Sibylle C Vonesch^{1 2}✉

Some other pros and cons of BE vs PE vs retrons vs HDR

<https://doi.org/10.1016/j.mib.2025.102705>

(and much more on precision editing)

focused on microbes but the general principles hold across organisms